



**Faculty of Engineering &
Technology
Electrical & Computer Engineering
Department**

**COMMUNICATION SYSTEMS-
ENEE3309**

Report Project phase#2

Prepared by:

Yara khattab 1210520

Saja Asfour 1210737

Baraa Nasar 1210880

Instructor: Dr. Qadri Mayyala

Section: 3 , 4

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Abstract:

This project involves the practical implementation of a circuit on a breadboard, serving as an amplitude modulation (AM) modulator and demodulator. The project aims to construct the circuit and test its functionality by generating modulated signals and subsequently demodulating them. The modulated and demodulated signals are visualized using an oscilloscope, with the results explained in detail. This implementation aims to provide an opportunity to interact with Amplitude modulation technology and understand the interaction of electrical components in a real-world circuit. Observing and explaining the results on the oscilloscope contributes to a comprehensive understanding of the circuit's performance in modulating and demodulating Amplitude-modulated signals, addressing practical challenges.

Table of Contents

Abstract:	1
Table of figures:	3
Results:	4
Part 1: Normal AM modulator	4
Part 1.1: Modulated single before BPF	4
Part 1.2: Modulated single after BPF	5
Part 2: Demodulated single	7
Part 3: Comparison between results	9
Conclusions:	10

Table of figures:

Figure 1: Circuit for $s(t)$ before BPF in Pspice	4
Figure 2: Simulation for $s(t)$ in time domain before BPF in Pspice	4
Figure 3: Bread-board circuit implementation and simulation for $s(t)$ before BPF.....	5
Figure 4: Circuit for $s(t)$ after BPF in Pspice	5
Figure 5: Simulation for $s(t)$ in time domain after BPF in Pspice	6
Figure 6: Bread-board circuit implementation and simulation for $s(t)$ after BPF.....	6
Figure 7: Circuit for demodulated signal in Pspice	7
Figure 8: Simulation for demodulated signal in Pspice	7
Figure 9: Bread-board circuit implementation and simulation for demodulated signal	8

Results:

Part 1: Normal AM modulator

Part 1.1: Modulated single before BPF

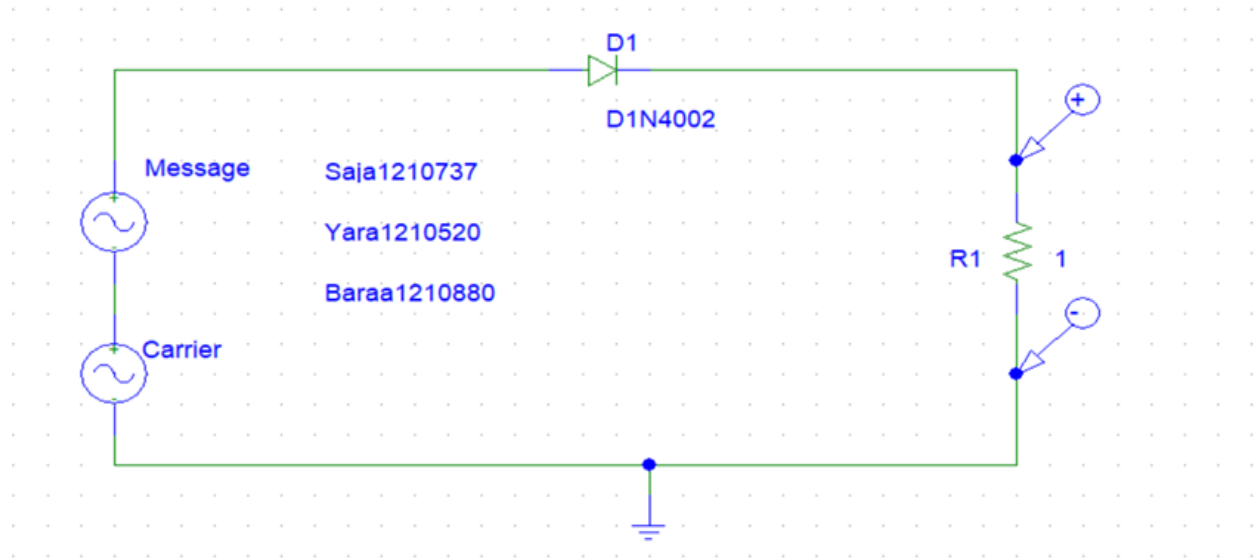


Figure 1: Circuit for $s(t)$ before BPF in Pspice

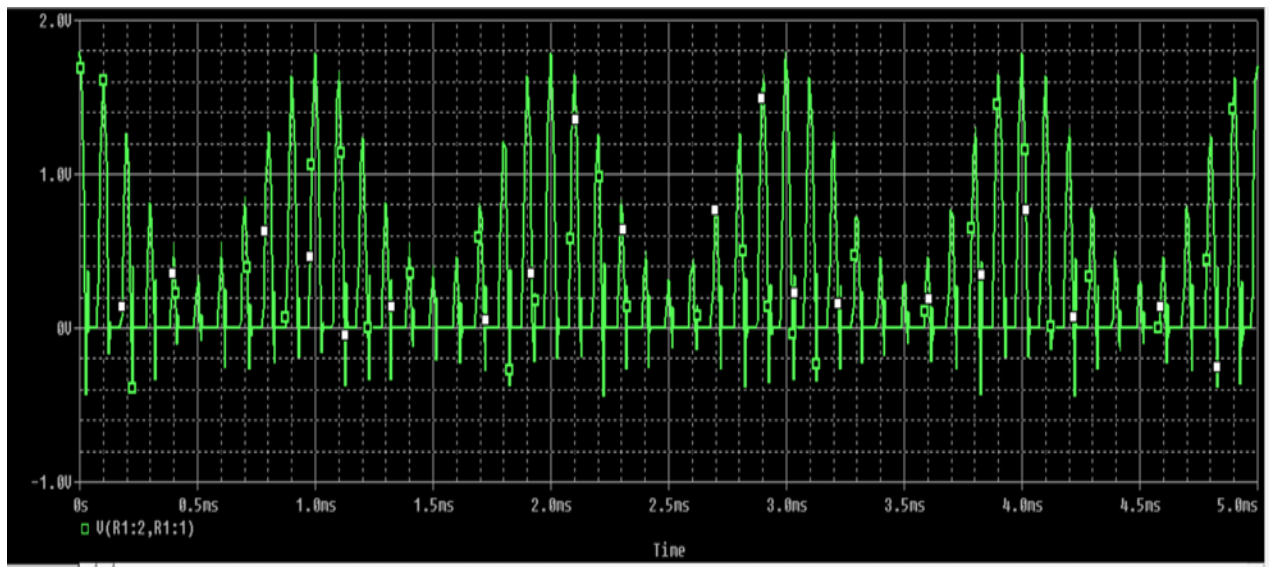


Figure 2: Simulation for $s(t)$ in time domain before BPF in Pspice

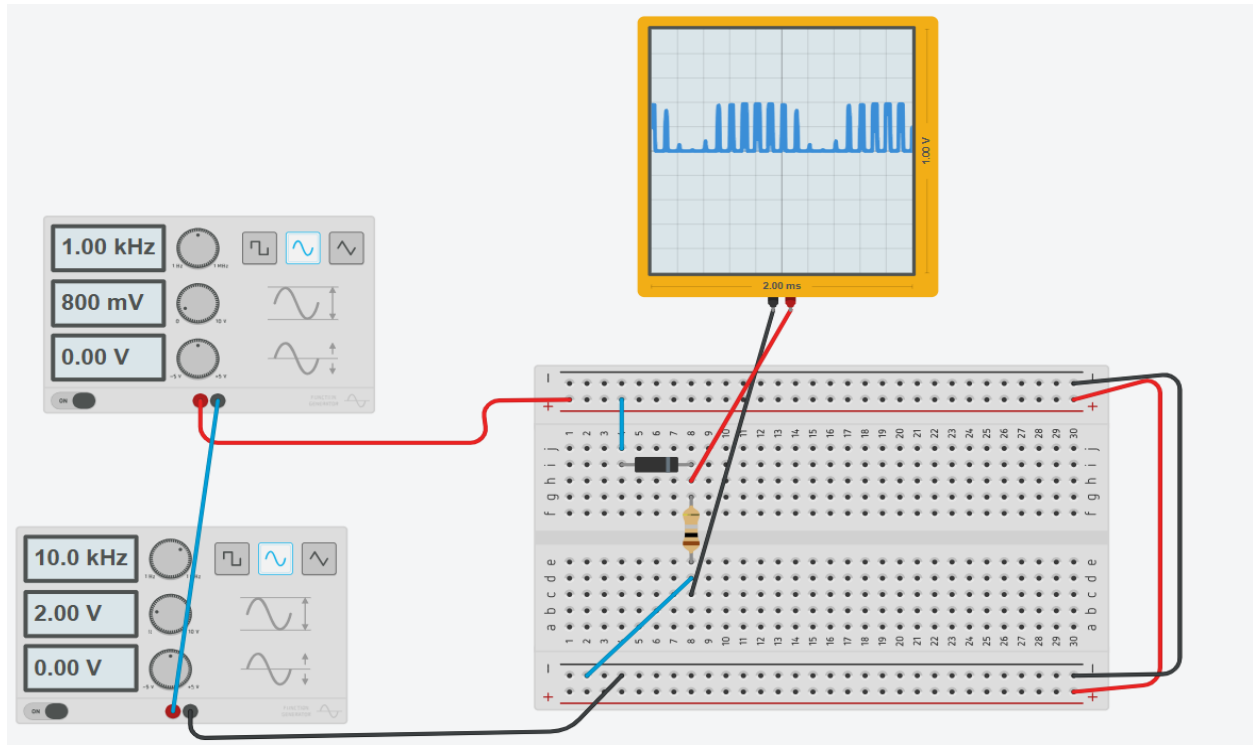


Figure 3: Bread-board circuit implementation and simulation for $s(t)$ before BPF

Part 1.2: Modulated single after BPF

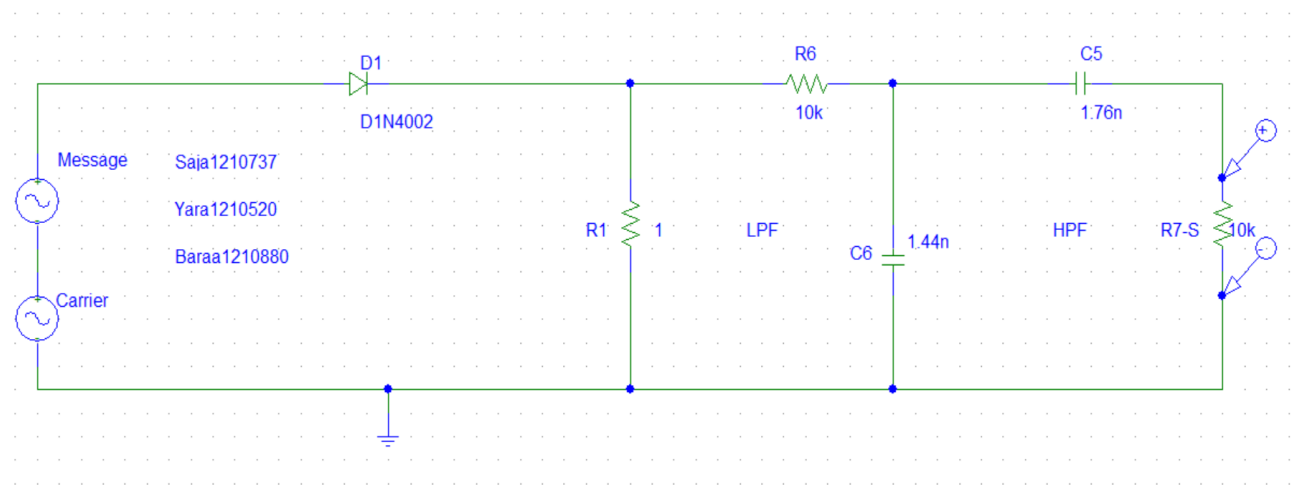


Figure 4: Circuit for $s(t)$ after BPF in Pspice



Figure 5: Simulation for $s(t)$ in time domain after BPF in Pspice

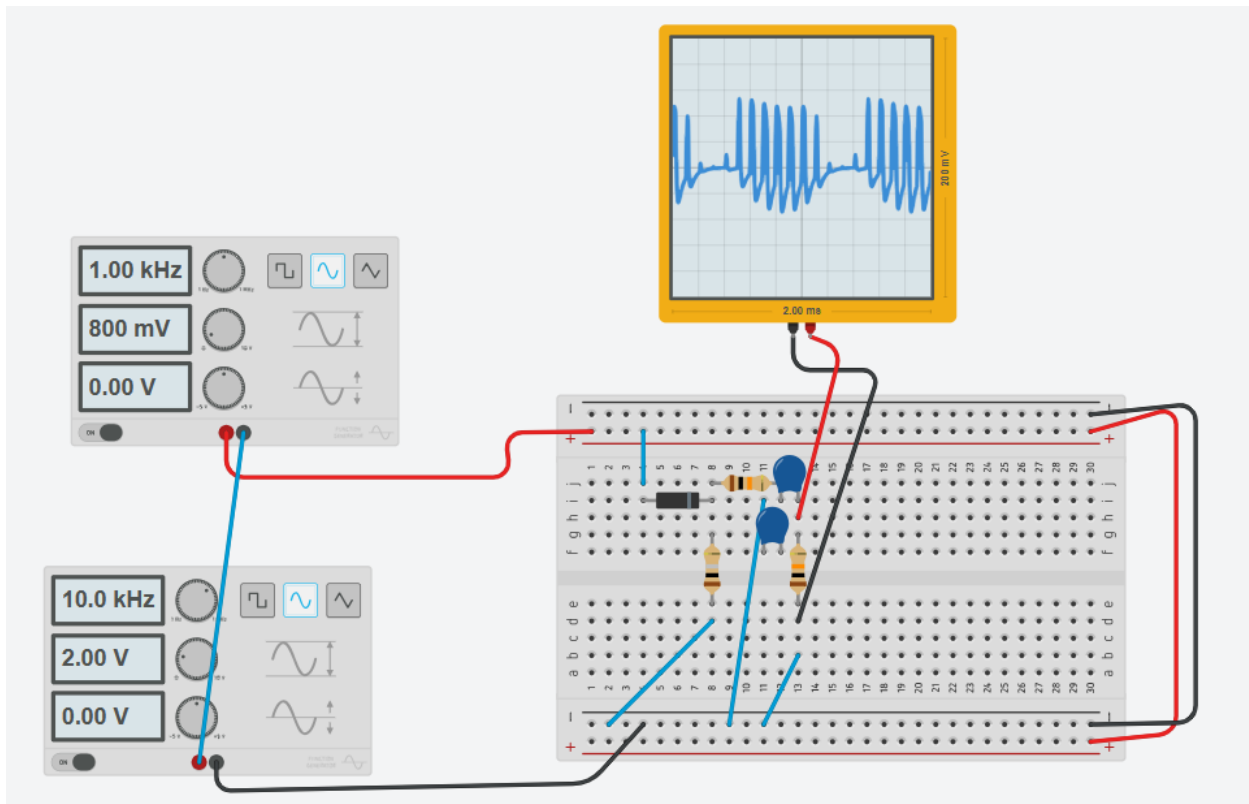


Figure 6: Bread-board circuit implementation and simulation for $s(t)$ after BPF

Part 2: Demodulated single

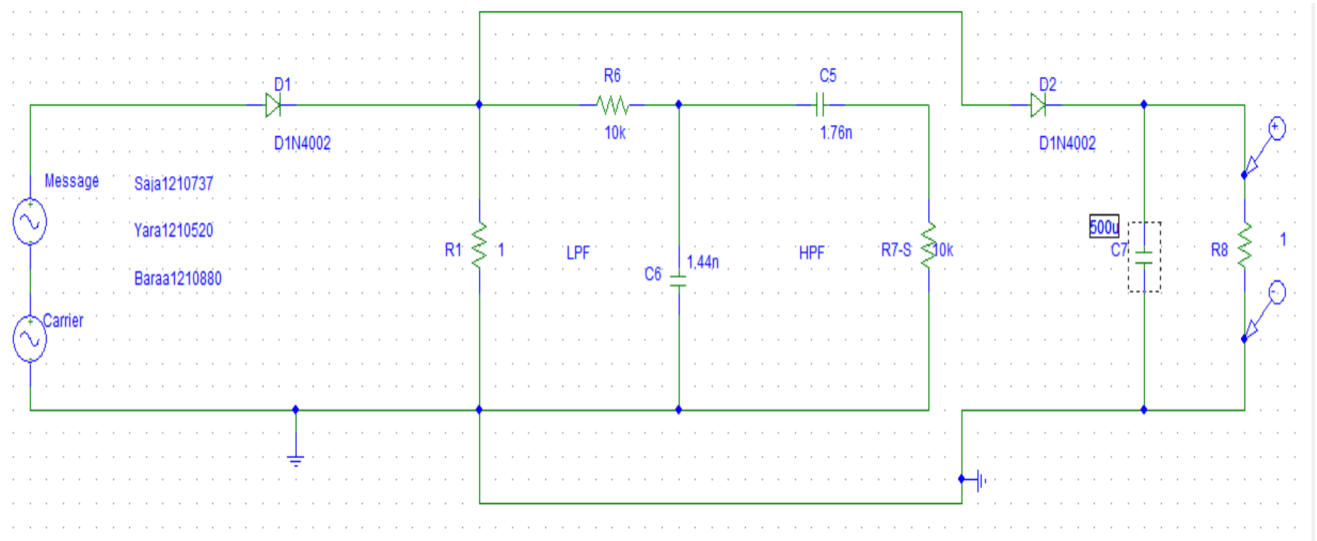


Figure 7: Circuit for demodulated signal in Pspice

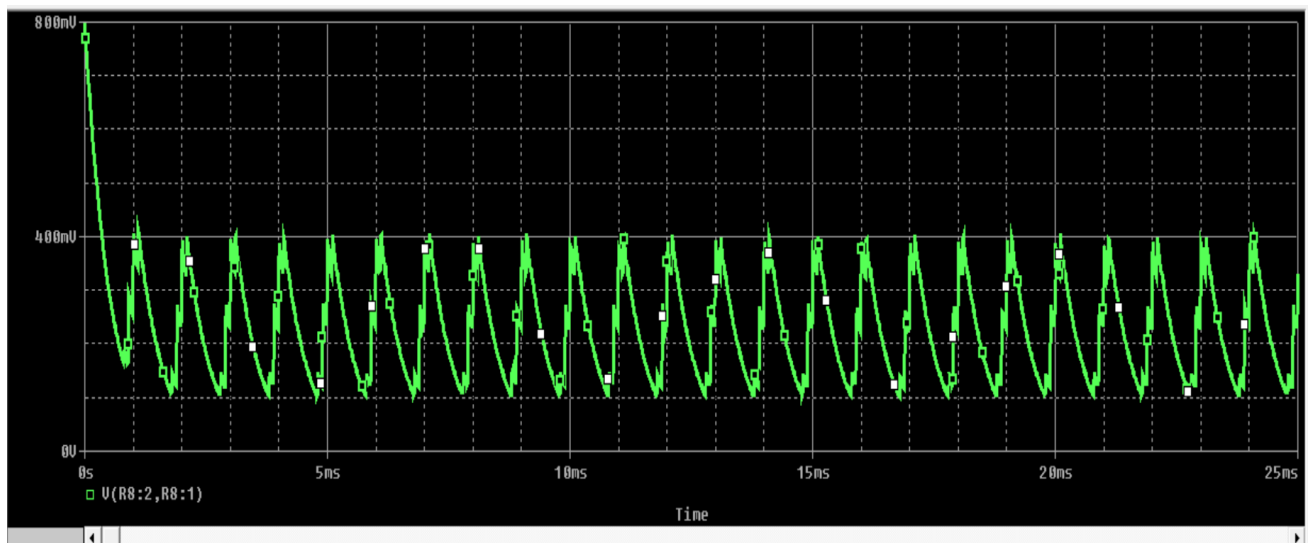


Figure 8: Simulation for demodulated signal in Pspice

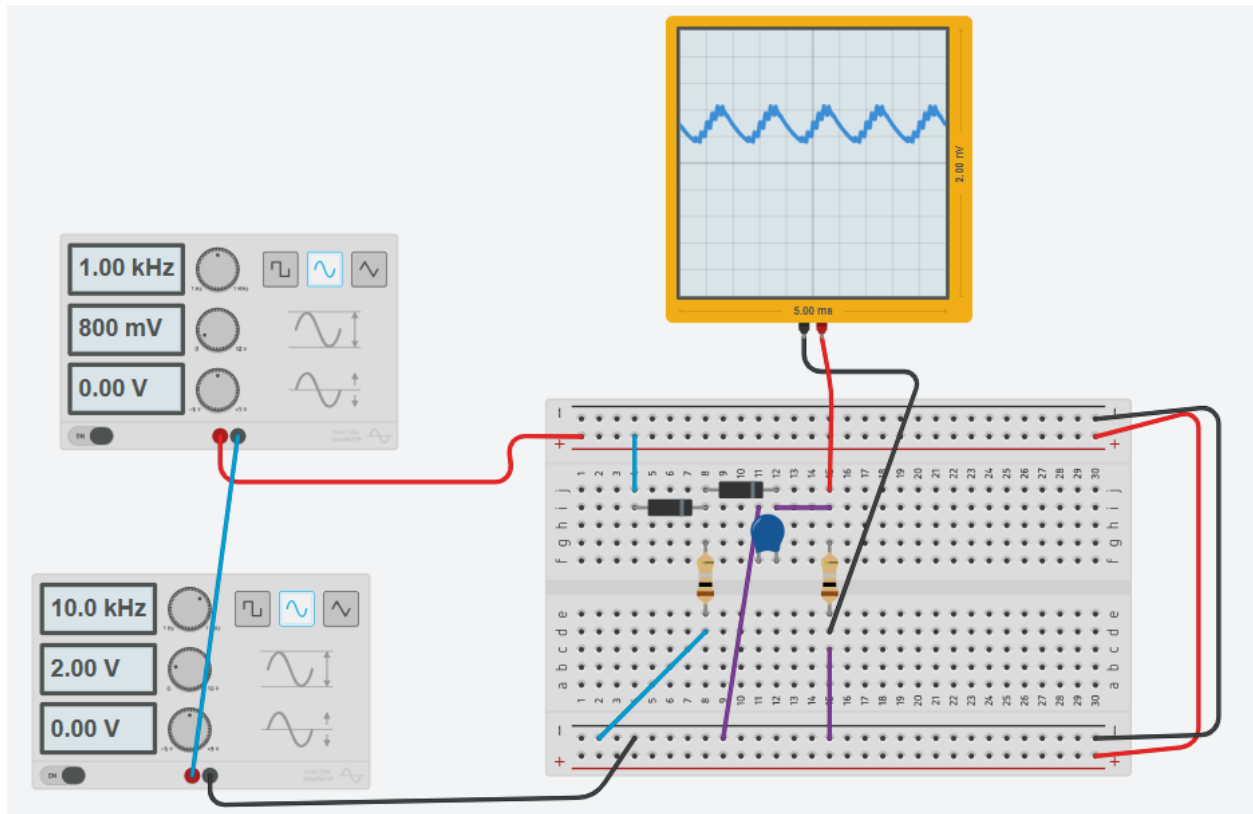


Figure 9: Bread-board circuit implementation and simulation for demodulated signal

Part 3: Comparison between results

In comparing the simulation results obtained from PSpice and Tinkercad for the given circuit, we ensured a consistent setup by replicating identical circuit configurations, component values, and initial conditions in both tools. Simulation parameters, including time step and convergence criteria, were carefully matched to establish a fair comparison.

The accuracy of component models was scrutinized to identify any discrepancies in the representation of resistors, capacitors. Transient analyses were conducted, and the resulting waveforms were compared for shape.

Also we noticed that the simulation of demodulated signal in Tinkercad closer to the message than the simulation in Pspice, maybe because we do not have an ideal diode in Pspice.

Additionally, despite efforts to maintain consistency, subtle variations were observed in certain simulation outputs, because we use cosine wave to make the message and carrier in Pspice but in Tinkercad we do not have cosine wave so we use sine wave in function generator which lead to phase shift between the two results.

Conclusions:

A significant success has been achieved in implementing the Amplitude Modulation (AM) circuit on the breadboard. An effective circuit for both modulation and demodulation was constructed, allowing us to successfully display modulated and demodulated signals on the oscilloscope. Through this project, practical understanding of AM modulation and demodulation processes has been enhanced, and the results have underscored the circuit's effectiveness and real-world application.